Dog Barking Analyzer

Product requirements document | Implementation handoff to Codex | Version 1.0

1. Problem and intended outcome

A dog is left alone in an apartment while the wife is away. A spare iPhone 14 running Voice Record Pro captures approximately two hours of continuous audio. The wife and the Windows desktop operator are currently on different continents. Reviewing the entire recording is tedious. The product must identify suspected dog vocalizations, preserve their original timestamps, and produce short clips and a clear review summary.

2. Agreed architecture

Record on iPhone -> upload to a private shared Google Drive folder -> manually download to Windows -> run local analysis -> review report and clips -> manually upload selected results to Google Drive for the wife.

Google Drive is the asynchronous exchange area. The two users need internet access, but do not need the same Wi-Fi network, a VPN, a running desktop at upload time, or a local shared folder. Audio classification occurs on the Windows computer. There is no cloud AI API or per-recording inference charge.

3. Scope and priorities

P0 / MVP: manual Drive exchange; Windows drag-and-drop launcher and CLI; FFmpeg decoding; local pretrained YAMNet classification; timestamped episodes; playable clips; local HTML report; CSV and JSON; portable ZIP for sharing; actionable failures and setup documentation.

P1 / later: optional local folder watching after the user configures Drive synchronization; user annotations; comparison across sessions. Direct Google Drive API authentication and remote job orchestration are separate future work.

Excluded: live notifications, video, remote microphone control, exact individual-bark counting, diagnosis of separation anxiety, identification of a particular dog, and automatic emails or messages.

4. Success criteria

A nontechnical user can process an exported recording without editing code. A two-hour recording finishes with complete timeline coverage or an explicit incomplete status. Review effort is measured as total exported clip duration, rather than assumed to always be a few minutes. No accuracy or processing-speed claim is made until measured on representative recordings.

Remote exchange and user flow

5. Google Drive operating procedure

The user creates a folder named Dog Audio Workspace, restricted to the two Google accounts, with Editor access where needed to upload files. Proposed subfolders: Incoming Audio, Results, and Reviewed. Available Drive storage and upload bandwidth constrain transfer; the solution must not require paid storage or publicly shared links.

Wife: stop and save the full recording; export the original M4A/AAC or MP3 to Incoming Audio using Voice Record Pro Google Drive export. If that integration cannot select the shared folder, export through the iOS share sheet / Files / Drive upload, or upload first and move it in Drive. Preserve the recording on the phone until the upload is confirmed and the file can be downloaded.

Operator: wait for the upload to finish, download the audio using Drive in a browser, and save it in a local input directory. Drag it onto Analyze Recording.bat or choose it through a file picker. This manual path is mandatory and must work without Drive for desktop or Google OAuth credentials.

After processing: review the report locally; upload the portable result ZIP and a plain-text summary to Results. The wife can read the summary and play separately uploaded MP3 clips in Drive, subject to Drive processing. For the full interactive report, download and extract the ZIP on a computer, then open index.html. Google Drive must not be treated as an HTML website host. iPhone HTML playback is not an MVP acceptance requirement.

6. Session naming and time zones

Suggest names such as 2026-10-07_dog-alone_session-01.m4a. Do not infer recording start time from upload time, filesystem modification time, or an ambiguous filename. Accept optional recording-start ISO 8601 with UTC offset and departure offset in seconds. The recording location may have a different time zone from the operator in Vancouver.

Always display elapsed HH:MM:SS from original audio start. Display wall-clock time only when explicitly supplied; include its offset. First bark after departure is unavailable unless departure offset is supplied. Mark recordings without reliable start metadata as elapsed-time-only.

7. Recording guidance

Record continuously; retain quiet periods; avoid silence-removal modes and speech-focused noise suppression. Place the phone safely with an unobstructed microphone and test a locked-screen recording. Upload the original compressed audio; WAV conversion happens locally. No audio needs to be recorded while people are present for the intended experiment.

Detection and metrics

8. Recommended analysis implementation

Use Python, FFmpeg/ffprobe, and pretrained YAMNet through a compatible TensorFlow runtime. Select and pin a tested 64-bit Windows/Python dependency combination during implementation. CPU execution is required; a GPU is optional and must not be necessary. Cache model assets locally after first download and verify subsequent offline analysis.

Validate the input with ffprobe. Decode mono 16 kHz PCM for analysis; normalize according to model expectations. Analyze all audio in bounded chunks, with overlap adequate for model context. Retain the original file and absolute timestamp offsets. Use the model's documented frame timing; explicitly account for padding and overlap. Do not average predictions across the entire recording.

Load class names from the model class map, not hard-coded numeric indices. Use Bark as the primary signal. Expose howling, yipping, whimpering and growling separately where the chosen model supports them; Dog is a broad supporting class and must not alone be presented as confirmed barking. Class scores are model scores, not calibrated probabilities.

9. Configurable event assembly

Initial engineering defaults, to be calibrated: onset score 0.30, offset score 0.20, merge gap 3 seconds, context padding 2 seconds before and after, analysis chunk 60 seconds. These values are proposals, not proven accuracy settings. Allow a single qualifying window to remain a candidate so isolated short barks are not automatically discarded.

Threshold qualifying frames using hysteresis, combine contiguous detections, then merge episodes separated by at most the configured gap. Clip boundaries are padded and clamped to recording bounds. Merge overlapping padded clips or split long clips with explicit episode references. Deduplicate overlap detections and never double-count covered time. Treat broad Dog-only candidates as a separate optional review category.

10. Metric definitions

Detected active time: union of qualifying model time intervals, capped at source duration; an approximate acoustic detection measure, not exact bark duration. Episode span: end minus start of a grouped event, including internal gaps. Review time: union of padded exported clip intervals. Report these separately.

Report recording duration, analyzed duration, suspected episode count, detected active time and percentage of analyzed duration, longest episode span, first detection, and review time. Never label episode count as bark count. An incomplete run must not show whole-recording conclusions. No detections means no detections at current settings, not proof of silence.

Deliverables and reliability

11. User-facing outputs

Create a unique result directory derived from a safe source stem and content hash. Include index.html, summary.txt, events.csv, results.json, clips/*.mp3, and an optional complete portable ZIP. Original audio is excluded from the ZIP by default. The HTML report is self-contained except for relative clip files, uses no remote CDN, and shows warnings, settings, timings and playable events.

Each event includes event ID, category, original start/end seconds and formatted elapsed times, peak and mean relevant scores, clip start/end, clip filename, and optional wall-clock times. CSV is UTF-8 with correct quoting. JSON additionally includes schema version, source SHA-256, source format/duration, model identity, dependency versions, run settings, analyzed intervals, processing duration, warnings and completion status.

summary.txt is suitable for reading directly after download on a phone. It must explain that events are suspected, show the key metrics and list clip filenames and original timestamps. Package relative paths so the report still works after extraction to a different Windows folder.

12. Windows interface and setup

Deliver setup.ps1, Analyze Recording.bat, a documented CLI, pinned dependency specification, sample configuration, README, and tests. Setup creates an isolated environment and checks FFmpeg and model availability. Use an officially sourced FFmpeg Windows build, document source/version/license, and verify downloads where checksums are available. Do not require administrator access when a per-user install is sufficient.

The launcher must accept spaces, Unicode and long filenames; open a file picker when no argument is passed; keep readable errors visible; show progress and current phase; and open the completed report. Example CLI contract: dog-audio analyze INPUT --output OUTPUT --config CONFIG. Reanalysis with changed settings produces a separate version or requires explicit overwrite.

13. Failure handling and safety of files

Reject missing, zero-byte, corrupt and unsupported input with actionable messages. Never delete or modify the source. Stage outputs in a temporary directory; publish complete outputs atomically where practical. On cancellation or crash, clean disposable files and retain a clear failure status. Detect insufficient disk space and prevent concurrent jobs from writing the same result directory.

Use subprocess argument arrays; never interpolate filenames into shell commands. Escape HTML and guard CSV cells against spreadsheet formula execution. Log errors without credentials. No telemetry or uploads by the analyzer. Network use is limited to explicit setup/model downloads in MVP.

Acceptance and implementation plan

14. Required acceptance tests

A. End to end: wife uploads a real Voice Record Pro file from a remote network; operator downloads and analyzes it on Windows; a shared result ZIP works after extraction on another computer. Independently verify the plain-text summary and standalone audio clips are usable by the wife.

B. Timeline: known synthetic events at file start/end and chunk boundaries retain source timestamps within one model hop plus documented frame-window uncertainty. Overlap processing adds no duplicate events. A two-hour file is fully covered with bounded memory; record runtime and peak memory on a documented reference desktop.

C. Robustness: test M4A/AAC, MP3 and WAV; spaces/Unicode paths; corrupt files; no detections; continuous noise; cancellation; low disk; missing FFmpeg; unavailable model download; cached offline analysis; portable report links; and repeated runs without unintended overwrite.

D. Detection calibration: annotate a representative sample from this apartment, including actual barking, quiet audio, doors, traffic, speech and any television. Tune on one subset and evaluate on a separate subset. Report event precision/recall using an explicit temporal matching rule; inspect unflagged segments for misses. Proposed pilot goal is at least 90% recall of annotated barking episodes; this is a goal, not a guarantee. If not met, expose the limitation and offer a more sensitive candidate-review mode.

E. Metrics: unit tests prove active-time union, episode spans and padded review time differ correctly. Missing start/departure metadata yields no invented wall-clock or after-departure values. Incomplete coverage cannot be reported as a complete session.

15. Delivery sequence and definition of done

1. Verify Windows dependency compatibility and model caching on a small sample. 2. Implement decode/classification with correct timing. 3. Add event grouping, clips and reports. 4. Add launcher, setup and error handling. 5. Validate two-hour operation and cross-continent Drive handoff. 6. Document calibration, limitations and user steps.

Done means a reproducible Windows installation, passing meaningful tests, a measured real-recording demo, portable outputs, and clear instructions requiring no code edits. A future watcher must wait for complete local files, avoid cloud placeholders and temporary downloads, deduplicate by content hash, and never move/delete Drive originals automatically.

16. Sources and implementation verification

Official references checked 7 October 2026. Recheck dependency and operating-system compatibility during implementation. Product defaults and acceptance targets above are design decisions, not claims from these sources.

Voice Record Pro export and recording features: https://www.bejbej.ca/app/voicerecordpro

Google Drive shared folders: https://support.google.com/drive/answer/7166529

YAMNet usage and input preparation: https://www.tensorflow.org/hub/tutorials/yamnet

YAMNet class map: https://github.com/tensorflow/models/blob/master/research/audioset/yamnet/yamnet_class_map.csv

FFmpeg filter reference: https://ffmpeg.org/ffmpeg-filters.html